ZERO-DIMENSIONAL COHERENT SHEAVES ARE COHOMOLOGICALLY TRIVIAL: A COHERENT SHEAF ON A LOCALLY NOETHERIAN SCHEME WHOSE SUPPORT HAS DIMENSION AT MOST 0 IS GENERATED BY GLOBAL SECTIONS AND HAS VANISHING HIGHER COHOMOLOGY.

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 2704-2710 of sources/stacks-coherent/pinned-original-source.tex; proof span is lines 2712-2727, the complete author proof (14 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Pool stacks-coherent backfill wave 33f.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. See why-field; every pointer named here occurs verbatim in the retained statement/proof text.

Source reference. Original lemma 01Y5: sources/stacks-coherent/pinned-original-source.tex, source lines 2627–2633. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01XB: sources/stacks-coherent/pinned-original-source.tex, source lines 363–367. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 089W: sources/stacks-coherent/pinned-original-source.tex, source lines 419–423. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0AAX: sources/stacks-coherent/pinned-original-source.tex, source lines 2727–2727. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let X be a locally Noetherian scheme. Let $\mathcal{F}$ be a coherent sheaf with $\dim(\mathit{Supp}(\mathcal{F})) \leq 0$. Then $\mathcal{F}$ is generated by global sections and $H^i(X, \mathcal{F}) = 0$ for $i > 0$.*

Date: stacks-project:coherent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. By Lemma 01Y5 we see that $\mathcal{F} = i_*\mathcal{G}$ where $i : Z \to X$ is the inclusion of the scheme theoretic support of $\mathcal{F}$ and where $\mathcal{G}$ is a coherent $\mathcal{O}_Z$-module. Since the dimension of Z is 0, we see Z is a disjoint union of affines (Properties, Lemma 0AAX). Hence $\mathcal{G}$ is globally generated and the higher cohomology groups of $\mathcal{G}$ are zero (Lemma 01XB). Hence $\mathcal{F} = i_*\mathcal{G}$ is globally generated. Since the cohomologies of $\mathcal{F}$ and $\mathcal{G}$ agree (Lemma 089W applies as a closed immersion is affine) we conclude that the higher cohomology groups of $\mathcal{F}$ are zero. $\square$